Top genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
CLDN16	0.2272932	274	3.356e−14	5.033e−10	5.239e−06
BPIFB1	0.1641271	334	1.355e−12	1.016e−08	5.289e−05
CLCNKA	0.1538276	312	4.089e−11	2.044e−07	7.093e−04
MAN2B1	0.1432193	321	1.117e−10	4.188e−07	8.923e−04
CYP11A1	0.1372012	329	1.690e−10	5.070e−07	8.923e−04
TREML2	0.1466091	302	2.615e−10	6.536e−07	8.923e−04
ADAM9	0.1345454	324	3.657e−10	6.856e−07	8.923e−04
CPB1	0.1288827	340	3.255e−10	6.856e−07	8.923e−04
SLC6A6	0.1423285	299	6.165e−10	1.027e−06	1.188e−03
MYH13	0.1434125	290	1.000e−09	1.500e−06	1.562e−03
CYP17A1	0.1180644	348	1.270e−09	1.732e−06	1.639e−03
TFB2M	0.1169758	337	2.919e−09	3.648e−06	3.138e−03
FGD5	0.1308761	299	3.396e−09	3.918e−06	3.138e−03
IL1R2	0.1126379	342	4.565e−09	4.891e−06	3.376e−03
CLPS	0.1290208	295	5.796e−09	5.513e−06	3.376e−03
TRPM6	0.1079474	351	6.249e−09	5.513e−06	3.376e−03
CEP192	0.1086965	349	6.111e−09	5.513e−06	3.376e−03
LRPPRC	0.1024708	362	9.275e−09	7.622e−06	4.176e−03
MAN2A1	0.1073003	345	9.656e−09	7.622e−06	4.176e−03
ADAM2	0.1160307	316	1.157e−08	8.676e−06	4.251e−03
CLCNKB	0.1237816	295	1.252e−08	8.941e−06	4.251e−03
EHHADH	0.1179007	306	1.554e−08	9.965e−06	4.251e−03
METAP1D	0.1270151	284	1.565e−08	9.965e−06	4.251e−03
TMEM131L	0.1034812	348	1.595e−08	9.965e−06	4.251e−03
BNC1	0.1127075	318	1.745e−08	1.047e−05	4.251e−03

Top genes by Q−Value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
CLDN16	0.2272932	274	3.356e−14	5.033e−10	5.239e−06
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TREML2	0.1466091	302	2.615e−10	6.536e−07	8.923e−04
ADAM9	0.1345454	324	3.657e−10	6.856e−07	8.923e−04
CPB1	0.1288827	340	3.255e−10	6.856e−07	8.923e−04
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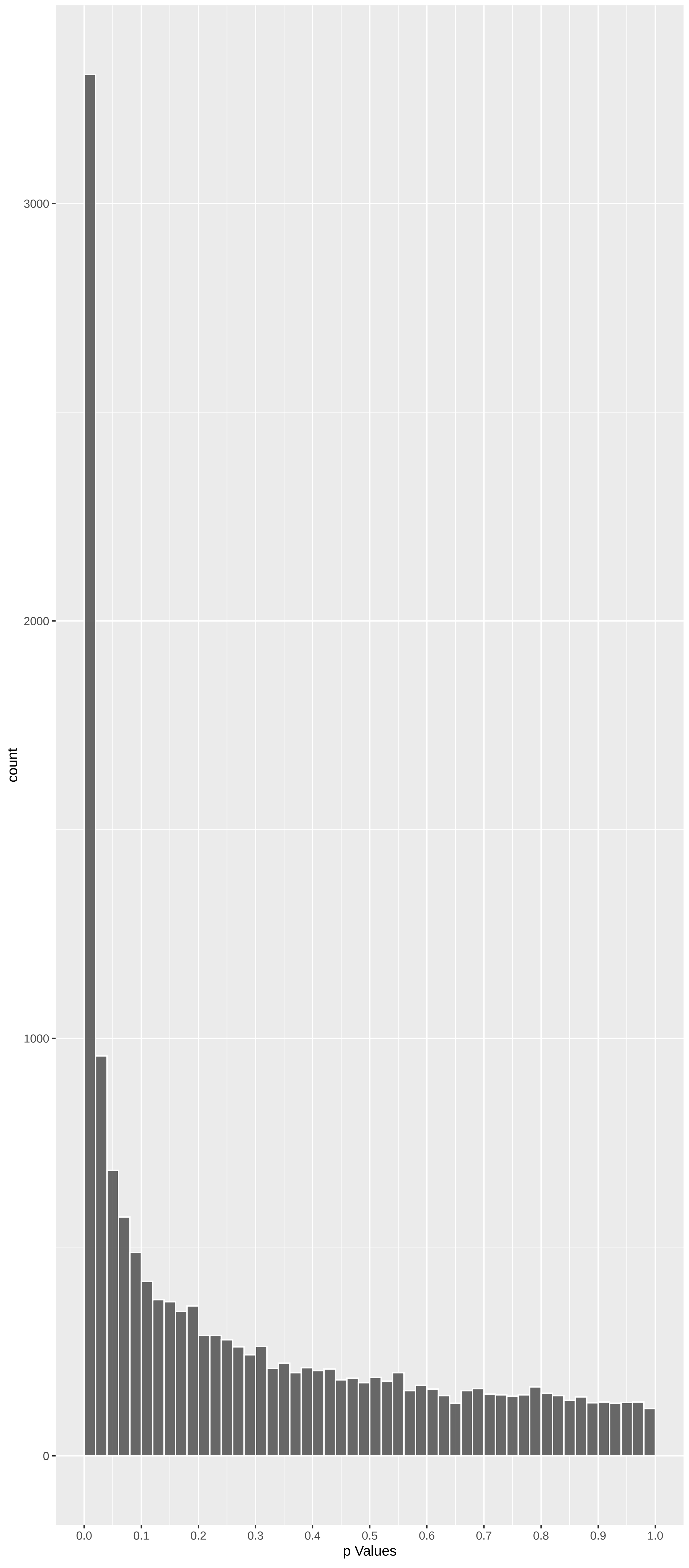
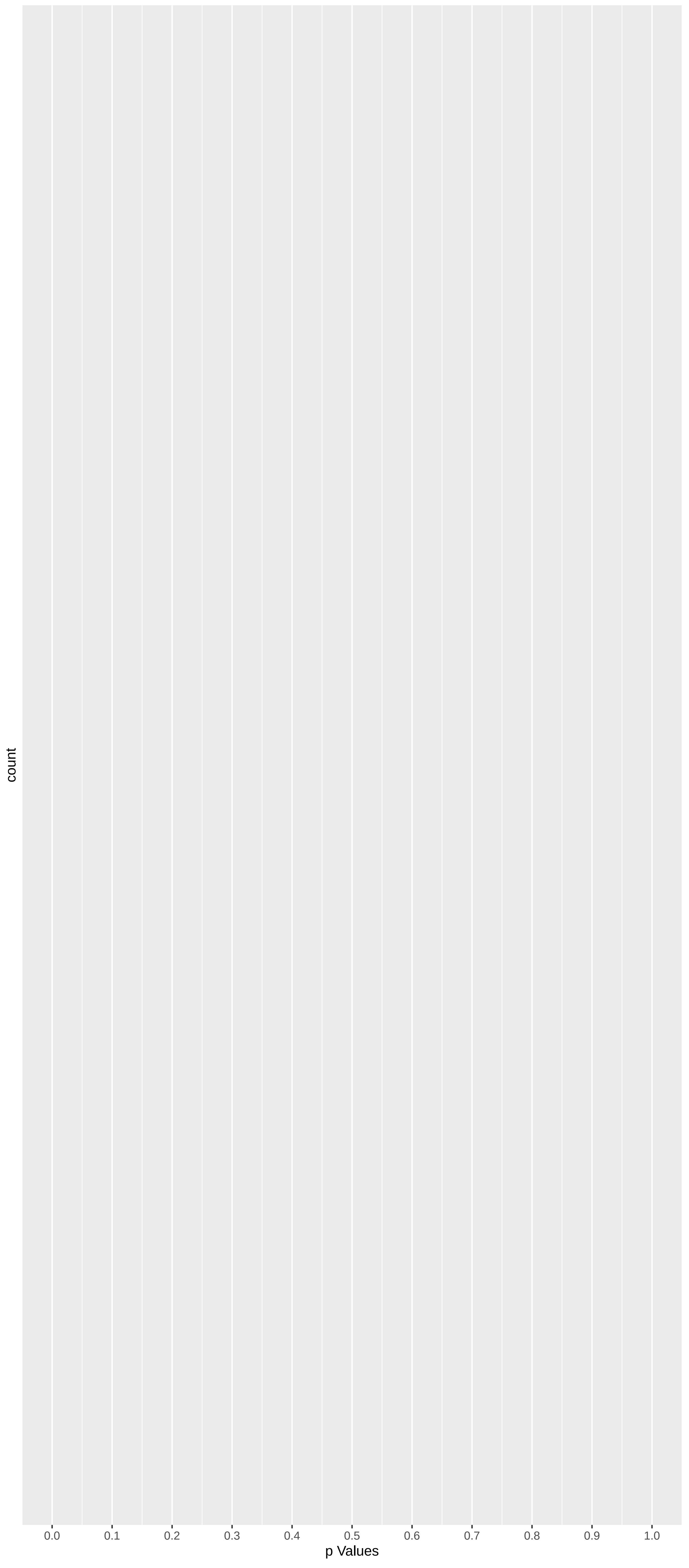
CYP17A1	0.1180644	348	1.270e−09	1.732e−06	1.639e−03
TFB2M	0.1169758	337	2.919e−09	3.648e−06	3.138e−03
FGD5	0.1308761	299	3.396e−09	3.918e−06	3.138e−03
CLPS	0.1290208	295	5.796e−09	5.513e−06	3.376e−03
IL1R2	0.1126379	342	4.565e−09	4.891e−06	3.376e−03
TRPM6	0.1079474	351	6.249e−09	5.513e−06	3.376e−03
CEP192	0.1086965	349	6.111e−09	5.513e−06	3.376e−03
LRPPRC	0.1024708	362	9.275e−09	7.622e−06	4.176e−03
MAN2A1	0.1073003	345	9.656e−09	7.622e−06	4.176e−03
EHHADH	0.1179007	306	1.554e−08	9.965e−06	4.251e−03
ADAM2	0.1160307	316	1.157e−08	8.676e−06	4.251e−03
METAP1D	0.1270151	284	1.565e−08	9.965e−06	4.251e−03
BNC1	0.1127075	318	1.745e−08	1.047e−05	4.251e−03
CLCNKB	0.1237816	295	1.252e−08	8.941e−06	4.251e−03
CPLX3	0.1799059	199	1.840e−08	1.062e−05	4.251e−03

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
CLDN16	0.2272932	274	3.356e−14	5.033e−10	5.239e−06
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CLCNKA	0.1538276	312	4.089e−11	2.044e−07	7.093e−04
MAN2B1	0.1432193	321	1.117e−10	4.188e−07	8.923e−04
CYP11A1	0.1372012	329	1.690e−10	5.070e−07	8.923e−04
TREML2	0.1466091	302	2.615e−10	6.536e−07	8.923e−04
ADAM9	0.1345454	324	3.657e−10	6.856e−07	8.923e−04
CPB1	0.1288827	340	3.255e−10	6.856e−07	8.923e−04
SLC6A6	0.1423285	299	6.165e−10	1.027e−06	1.188e−03
MYH13	0.1434125	290	1.000e−09	1.500e−06	1.562e−03

Top Negative genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Monocarboxylic Acid Transport (GO:001571	0.17370502	61	2.770e−06	1.492e−02	SLC16A5:36 SLCO1B1:38 SLC5A12:96 SLCO1A2:128 SLC22A13:218 SLC6A12:240
Regulation Of Cell Population Proliferat	−0.04988025	627	2.571e−05	6.922e−02	AVPR2:7 SMARCC1:15 MAPK15:34 ERBB4:40 HAS2:93 TSPAN31:97
T Cell Differentiation In Thymus (GO:003	−0.31699960	13	7.591e−05	6.934e−02	PTPRC:30 RIPK3:631 ZFP36L2:644 ZFP36L1:805 SHH:1287 LIG4:1446
Bile Acid And Bile Salt Transport (GO:00	0.28702848	16	7.059e−05	6.934e−02	SLCO1B1:38 SLCO1A2:128 SLC27A5:257 SLC10A2:329 SLC51A:763 SLC10A3:946
Cytoplasmic Translation (GO:0002181)	0.17043577	45	7.726e−05	6.934e−02	UBA52:310 FTSJ1:319 RPS27A:489 RPS18:509 RPL34:537 RPS15A:712
Organic Hydroxy Compound Transport (GO:0	0.19417580	35	7.089e−05	6.934e−02	SLCO1B1:38 SLCO1A2:128 ABCG8:232 SLC27A5:257 SLC10A2:329 RALBP1:455
Positive Regulation Of DNA−templated Tra	−0.03581660	1024	1.496e−04	1.151e−01	E2F8:6 SMARCC1:15 PPARGC1A:21 ERBB4:40 EP300:46 HMGN1:52
Positive Regulation Of Cell Population P	−0.05615917	383	1.811e−04	1.219e−01	IL5RA:3 SMARCC1:15 PTPRC:30 MAPK15:34 ERBB4:40 HAS2:93
piRNA Processing (GO:0034587)	0.25739148	17	2.393e−04	1.275e−01	PIWIL1:33 GPAT2:150 MOV10L1:170 ANKRD34C:384 ASZ1:839 DDX4:883
Positive Regulation Of Transcription By	−0.03938354	767	2.605e−04	1.275e−01	E2F8:6 PPARGC1A:21 EP300:46 GLIS3:54 NKX2−1:58 MMP12:64
Positive Regulation Of Transcription Reg	−0.26646546	16	2.246e−04	1.275e−01	EP300:46 DDRGK1:240 RB1:433 H1−0:663 NEUROD1:844 PAX6:941
Positive Regulation Of Cellular Process	−0.04859186	469	3.555e−04	1.497e−01	SMARCC1:15 MAPK15:34 ERBB4:40 CITED2:68 CCR7:75 HAS2:93
Skin Development (GO:0043588)	0.14311359	52	3.613e−04	1.497e−01	ADAM9:8 ACER1:117 OPN3:534 ITGB4:587 ST14:849 EVPL:1107
U4 snRNA 3'−End Processing (GO:0034475)	−0.37094163	7	6.774e−04	1.967e−01	EXOSC5:532 EXOSC7:1060 EXOSC8:1189 EXOSC9:1229 EXOSC3:1334 EXOSC6:1709
Cellular Response To Growth Factor Stimu	−0.08431641	135	7.413e−04	1.967e−01	SEMA6A:90 HAS2:93 NTRK1:102 ANKRD1:248 CORO1B:256 SPRY2:346
Detection Of Chemical Stimulus Involved	−0.14597299	47	5.411e−04	1.967e−01	OR2T1:41 OR10G3:291 OR10K2:319 OR10T2:466 OR2S2:679 OR10G6:722
Detection Of Chemical Stimulus Involved	−0.14637207	46	5.991e−04	1.967e−01	OR2T1:41 OR10G3:291 OR10K2:319 OR10T2:466 OR2S2:679 OR10G6:722
Detection Of Chemical Stimulus Involved	0.20505737	22	8.732e−04	1.967e−01	GNAT2:588 PKD1L3:619 TAS2R41:657 TAS2R10:801 TAS2R60:897 GNAT3:933
Hepatocyte Differentiation (GO:0070365)	−0.43463817	5	7.631e−04	1.967e−01	E2F8:6 E2F7:646 PCK2:1139 PROX1:1214 PCK1:1803 NA
Keratinocyte Differentiation (GO:0030216	0.18507847	27	8.766e−04	1.967e−01	ADAM9:8 OPN3:534 ST14:849 EVPL:1107 TGM1:1988 EXPH5:2540
Negative Regulation Of Blood Pressure (G	0.26716909	13	8.532e−04	1.967e−01	DRD2:1074 DRD3:1214 PRCP:1346 GPR37L1:2023 NOS1:2624 RNLS:2852
Nuclear−Transcribed mRNA Catabolic Proce	−0.37094163	7	6.774e−04	1.967e−01	EXOSC5:532 EXOSC7:1060 EXOSC8:1189 EXOSC9:1229 EXOSC3:1334 EXOSC6:1709
Regulation Of ERK1 And ERK2 Cascade (GO:	−0.06942455	195	8.686e−04	1.967e−01	NEK10:23 PTPRC:30 ERBB4:40 CCR7:75 CCR1:79 SEMA6A:90
Sensory Perception Of Bitter Taste (GO:0	0.22889422	19	5.539e−04	1.967e−01	GNAT2:588 TAS2R41:657 TAS2R10:801 CALHM1:860 TAS2R60:897 GNAT3:933
NADH Dehydrogenase Complex Assembly (GO:	0.13669295	48	1.061e−03	1.997e−01	TMEM126A:101 NDUFS5:196 ECSIT:253 NDUFB8:325 NDUFS3:356 NDUFA9:440
Acid Secretion (GO:0046717)	0.33417053	8	1.064e−03	1.997e−01	SLC22A16:70 UGT1A6:604 SLC51A:763 DRD3:1214 SLC51B:2740 SLC26A6:3639
Amino Acid Transmembrane Transport (GO:0	0.14701039	41	1.134e−03	1.997e−01	SLC6A6:9 SLC22A4:668 SLC7A9:774 SLC38A4:1193 SLC7A10:1362 SLC36A2:1506
Mitochondrial Respiratory Chain Complex	0.13669295	48	1.061e−03	1.997e−01	TMEM126A:101 NDUFS5:196 ECSIT:253 NDUFB8:325 NDUFS3:356 NDUFA9:440
Phospholipid Translocation (GO:0045332)	0.18083229	27	1.150e−03	1.997e−01	ATP9B:332 ANO6:514 ABCA4:997 ABCB1:1466 ATP8B2:1903 ATP11A:1917
Regulation Of Organelle Organization (GO	−0.10394287	84	1.009e−03	1.997e−01	TOM1:17 MAPK15:34 EP300:46 PINK1:50 RAP1GDS1:67 MAPK3:205
Steroid Hormone Biosynthetic Process (GO	0.23541953	16	1.115e−03	1.997e−01	CYP11A1:5 CYP17A1:11 CYP21A2:277 HSD17B2:390 HSD17B6:846 HSD17B3:1140
Endoplasmic Reticulum Tubular Network Or	−0.24847815	14	1.288e−03	2.109e−01	TMEM33:378 RTN1:507 RETREG3:509 RTN4:513 ATL1:752 ZFYVE27:1185
Positive Regulation Of ERK1 And ERK2 Cas	−0.07870628	141	1.292e−03	2.109e−01	PTPRC:30 ERBB4:40 CCR7:75 CCR1:79 NTRK1:102 MAPK3:205
Detection Of Chemical Stimulus Involved	0.21695600	18	1.443e−03	2.230e−01	GNAT2:588 TAS2R41:657 TAS2R10:801 TAS2R60:897 GNAT3:933 TAS2R1:2267
Inorganic Cation Import Across Plasma Me	0.09571436	93	1.450e−03	2.230e−01	SCNN1G:168 KCNJ1:188 ATP2B4:403 SCNN1B:469 TRPM1:612 SCN11A:681
Natural Killer Cell Differentiation (GO:	−0.40929694	5	1.527e−03	2.283e−01	PTPRC:30 PIK3CD:624 RABL3:1717 NFIL3:1936 TOX:2378 NA
Amino−Acid Betaine Transport (GO:0015838	0.34206847	7	1.724e−03	2.336e−01	SLC22A16:70 SLC6A12:240 SLC22A4:668 SLC6A20:1870 SLC7A6:2117 SLC22A5:3732
Circulatory System Development (GO:00723	−0.08553614	113	1.722e−03	2.336e−01	ERBB4:40 CITED2:68 PDLIM1:198 POU6F1:238 DNAAF4:306 HSPG2:313
Intracellular Monoatomic Ion Homeostasis	0.20670617	19	1.818e−03	2.336e−01	CLDN16:1 RHCG:91 CNNM4:194 RHAG:313 SLC4A1:625 LETMD1:1363
Neuron Differentiation (GO:0030182)	−0.07573388	143	1.822e−03	2.336e−01	ERBB4:40 WNT9B:92 BRSK1:107 VAX1:164 FZD8:201 SPINK5:219

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HOUNKPE_HOUSEKEEPING_GENES	−0.04368203	853	1.798e−05	3.798e−02	SNRNP70:4 PPP2R5A:10 SMARCC1:15 GPBP1:71 TAOK2:74 SCYL1:81
MILI_PSEUDOPODIA_HAPTOTAXIS_UP	−0.06138209	407	2.342e−05	3.798e−02	ARHGAP5:44 TRAPPC8:47 GPBP1:71 RIOK3:80 SCP2:91 TUT7:142
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.05052541	624	1.887e−05	3.798e−02	CLCNKA:3 SLC6A6:9 TRPM6:17 CLCNKB:21 ANO5:31 SLCO1B1:38
HSIAO_LIVER_SPECIFIC_GENES	0.08937665	196	1.668e−05	3.798e−02	AGXT:27 ATP13A3:40 ALDH1A1:127 SORD:165 PROC:171 CYP27A1:231
REACTOME_SARS_COV_2_MODULATES_HOST_TRANS	0.21602218	31	3.156e−05	4.094e−02	SNRPD3:375 RPS27A:489 RPS18:509 SNRPD1:656 RPS15A:712 RPS23:735
MIKKELSEN_MEF_HCP_WITH_H3_UNMETHYLATED	0.08645661	186	4.941e−05	4.579e−02	TRPM6:17 PIWIL1:33 TLE6:61 FAR2:66 NSD1:76 KCNV2:142
FERREIRA_EWINGS_SARCOMA_UNSTABLE_VS_STAB	0.10098979	136	4.893e−05	4.579e−02	FANCM:156 CHAF1A:341 SNRPD3:375 IKBKB:432 NDUFA9:440 PINX1:448
KIM_RESPONSE_TO_TSA_AND_DECITABINE_UP	0.11211014	105	7.337e−05	5.949e−02	IL1R2:14 KCNV2:142 GDF15:569 IL1RAP:594 STAG3:643 SMPD3:724
BLALOCK_ALZHEIMERS_DISEASE_DN	−0.03710541	996	9.002e−05	6.488e−02	ARHGEF4:36 PINK1:50 MRPL20:59 DGUOK:66 RAP1GDS1:67 SCP2:91
REACTOME_PROCESSING_OF_DNA_DOUBLE_STRAND	0.13414338	70	1.051e−04	6.821e−02	RNF168:78 UBA52:310 CLSPN:335 H2BC21:461 RPS27A:489 BRCA1:501
REACTOME_OLFACTORY_SIGNALING_PATHWAY	−0.11257732	93	1.777e−04	8.302e−02	OR2T1:41 OR10G3:291 OR10K2:319 OR10T2:466 OR52B6:567 OR52E2:657
REACTOME_BUTYRATE_RESPONSE_FACTOR_1_BRF1	−0.26939478	16	1.910e−04	8.302e−02	XRN1:148 EXOSC5:532 ZFP36L1:805 DCP1A:1017 EXOSC7:1060 EXOSC8:1189
WP_DRUG_INDUCTION_OF_BILE_ACID_PATHWAY	0.30249750	13	1.591e−04	8.302e−02	ABCB11:35 SLCO1B1:38 CYP7A1:690 SLC51A:763 NR1H4:776 SLC10A1:1356
KIM_ALL_DISORDERS_CALB1_CORR_UP	−0.05221443	439	1.897e−04	8.302e−02	SNX17:14 UBQLN2:28 PINK1:50 AMPH:161 WDR47:228 FBXO9:277
DUTERTRE_ESTRADIOL_RESPONSE_24HR_UP	0.06342925	294	1.920e−04	8.302e−02	IQGAP3:86 KCNK15:92 DNMT1:105 ESCO2:113 UHRF1:176 HAUS8:185
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.43316026	6	2.382e−04	9.182e−02	CA4:162 RHAG:313 CA1:496 SLC4A1:625 AQP1:1539 CA2:2933
REACTOME_SPERM_MOTILITY_AND_TAXES	0.39856713	7	2.603e−04	9.182e−02	KCNU1:81 CATSPERG:183 CATSPERB:210 CATSPERD:423 CATSPER4:622 CATSPER3:3252
REACTOME_ION_CHANNEL_TRANSPORT	0.08190641	166	2.783e−04	9.182e−02	CLCNKA:3 TRPM6:17 CLCNKB:21 ANO5:31 ASIC5:89 TRPM5:109
WP_CYTOPLASMIC_RIBOSOMAL_PROTEINS	0.15646892	45	2.831e−04	9.182e−02	UBA52:310 RPS27A:489 RPS18:509 RPL34:537 RPS15A:712 RPS23:735
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20024192	28	2.455e−04	9.182e−02	CYP11A1:5 CYP17A1:11 SULT1E1:44 CYP1A1:87 COMT:122 CYP21A2:277
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.15218101	47	3.084e−04	9.525e−02	NDUFS5:196 ECSIT:253 NDUFB8:325 NDUFS3:356 NDUFB5:502 NDUFB4:578
REACTOME_STIMULI_SENSING_CHANNELS	0.10156946	102	3.987e−04	9.578e−02	CLCNKA:3 TRPM6:17 CLCNKB:21 ANO5:31 ASIC5:89 TRPM5:109
REACTOME_TRISTETRAPROLIN_TTP_ZFP36_BINDS	−0.26463404	15	3.874e−04	9.578e−02	XRN1:148 EXOSC5:532 DCP1A:1017 EXOSC7:1060 EXOSC8:1189 EXOSC9:1229
REACTOME_PIWI_INTERACTING_RNA_PIRNA_BIOG	0.24246914	18	3.691e−04	9.578e−02	PIWIL1:33 MOV10L1:170 PIWIL2:239 POLR2J:417 POLR2I:560 ASZ1:839
BRUINS_UVC_RESPONSE_VIA_TP53_GROUP_B	−0.04742304	484	3.798e−04	9.578e−02	IL5RA:3 TRIM3:101 CCDC127:152 CEP85L:196 COL4A1:209 QSOX1:212
WEBER_METHYLATED_HCP_IN_FIBROBLAST_DN	0.17580017	34	3.900e−04	9.578e−02	LPIN1:125 MOV10L1:170 DKKL1:323 BOLL:942 SYCP2:1417 TRPM4:1455
WINTER_HYPOXIA_METAGENE	−0.07127746	210	3.820e−04	9.578e−02	PLOD2:37 ARHGAP5:44 CITED2:68 RIOK3:80 MAP4:150 TPI1:171
FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN	0.07749059	172	4.659e−04	9.636e−02	ANO5:31 SMC4:75 IQGAP3:86 ESCO2:113 MIS18BP1:119 KIF18B:135
MEISSNER_NPC_HCP_WITH_H3_UNMETHYLATED	0.04715317	478	4.442e−04	9.636e−02	TRPM6:17 ANO5:31 FAR2:66 RHCG:91 TMPRSS2:110 NAA11:157
REACTOME_REPRODUCTION	0.10473207	95	4.242e−04	9.636e−02	ADAM2:20 KCNU1:81 SYNE1:100 B4GALT1:129 CATSPERG:183 CATSPERB:210
REACTOME_RECYCLING_OF_BILE_ACIDS_AND_SAL	0.25190572	16	4.858e−04	9.636e−02	ABCB11:35 SLCO1B1:38 SLCO1A2:128 SLC27A5:257 ALB:299 SLC10A2:329
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.18151895	31	4.702e−04	9.636e−02	SLC6A6:9 SLC6A12:240 SLC43A1:737 SLC7A9:774 SLC6A19:1179 SLC38A4:1193
BIOCARTA_INTRINSIC_PATHWAY	−0.21976669	21	4.902e−04	9.636e−02	COL4A1:209 F9:443 COL4A2:607 F2:777 F2R:802 F8:1176
CHANGOLKAR_H2AFY_TARGETS_UP	0.14888645	45	5.516e−04	1.022e−01	SCD:189 CA1:496 SERPINA7:548 RORA:586 GLO1:618 ENPP3:644
REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR	−0.07822108	165	5.385e−04	1.022e−01	ACKR2:2 AVPR2:7 HEBP1:60 CCR7:75 CCR1:79 NPW:151
REACTOME_GENE_SILENCING_BY_RNA	0.10821380	85	5.691e−04	1.026e−01	PIWIL1:33 MOV10L1:170 PIWIL2:239 POLR2J:417 H2BC21:461 POLR2I:560
REACTOME_CARGO_CONCENTRATION_IN_THE_ER	−0.19060542	27	6.089e−04	1.068e−01	TMED10:220 SEC24C:708 LMAN2:859 F8:1176 LMAN2L:1221 STX5:1959
KEGG_TASTE_TRANSDUCTION	0.16631591	35	6.634e−04	1.133e−01	TRPM5:109 SCNN1G:168 SCNN1B:469 TAS1R1:638 TAS2R41:657 TAS2R10:801
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	0.14940108	43	7.024e−04	1.168e−01	EHHADH:22 BCAT2:47 ACAD8:314 ACADM:590 IL4I1:809 HIBCH:952
REACTOME_VITAMIN_D_CALCIFEROL_METABOLISM	0.30632133	10	7.956e−04	1.259e−01	CUBN:201 UBE2I:507 LGMN:1629 CYP2R1:2123 VDR:2625 PIAS4:2982

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Renal salt wasting	0.36698975	15	8.637e−07	8.469e−03	CLCNKA:3 CYP11A1:5 CLCNKB:21 SCNN1G:168 KCNJ1:188 CYP21A2:277
Renal tubular acidosis	0.20636568	35	2.412e−05	1.183e−01	CLDN16:1 TRPM6:17 CLCNKB:21 EHHADH:22 KCNJ1:188 MRGPRF:268
Hyperkalemia	0.27341192	17	9.533e−05	2.991e−01	CYP11A1:5 CYP17A1:11 LPIN1:125 SCNN1G:168 SCNN1B:469 WNK4:831
Nephrocalcinosis	0.15414757	52	1.220e−04	2.991e−01	CLDN16:1 TRPM6:17 CLCNKB:21 AGXT:27 NSD1:76 SLC26A3:172
Hypochloremia (disorder)	0.43763046	6	2.054e−04	3.356e−01	CLCNKA:3 CLCNKB:21 SLC26A3:172 KCNJ1:188 BSND:1259 SLC12A1:3855
Serum chloride level decreased (finding)	0.43763046	6	2.054e−04	3.356e−01	CLCNKA:3 CLCNKB:21 SLC26A3:172 KCNJ1:188 BSND:1259 SLC12A1:3855
Hyperactive renin−angiotensin system	0.36970772	8	2.934e−04	3.597e−01	CLCNKB:21 SCNN1G:168 SLC26A3:172 KCNJ1:188 SCNN1B:469 SCNN1A:2227
Increased plasma renin activity	0.36970772	8	2.934e−04	3.597e−01	CLCNKB:21 SCNN1G:168 SLC26A3:172 KCNJ1:188 SCNN1B:469 SCNN1A:2227
Cholecystolithiasis	0.13037476	61	4.345e−04	4.734e−01	CYP17A1:11 ABCB11:35 SLCO1B1:38 CYP1A1:87 SLCO1A2:128 ABCG8:232
Atrophy, Disuse	−0.36593425	7	8.002e−04	5.276e−01	PPARGC1A:21 HSPA4:914 PITX1:1442 PPARGC1B:1907 LIF:1964 FBXO32:3063
Abnormality of macular pigmentation	0.20394742	21	1.218e−03	5.276e−01	PROM1:41 GPR179:281 TRPM1:612 PDE6B:973 ABCA4:997 GRM6:1229
Abnormality of the urinary system	0.20366478	21	1.237e−03	5.276e−01	GLB1:305 HGD:566 RPGRIP1L:703 COL3A1:1451 RPS27:1555 RPS19:1607
Acute kidney injury	0.11240310	72	9.881e−04	5.276e−01	A2M:77 HEXB:124 LPIN1:125 ALB:299 AFM:415 AMBP:678
Benign recurrent intrahepatic cholestasi	0.38501572	6	1.091e−03	5.276e−01	ABCB11:35 GGTLC3:198 GGTLC1:851 GGT2:1819 GGT1:2029 ATP8B1:5230
Complicated pneumoconiosis	0.32293117	9	7.948e−04	5.276e−01	SLCO6A1:132 GSTK1:544 IL6:778 VEGFA:1734 ICAM1:1981 IL1A:2904
Decreased glomerular filtration rate	0.35590228	7	1.111e−03	5.276e−01	CLCNKA:3 CLCNKB:21 AGXT:27 BSND:1259 G6PC:2694 ITGA3:2798
Generalized glycogen storage disease of	0.22966542	17	1.046e−03	5.276e−01	SI:83 UGT1A6:604 GLA:865 ARSB:1106 DES:2681 G6PC:2694
Glycinuria	0.27374205	13	6.331e−04	5.276e−01	ACADM:590 SLC6A19:1179 GCSH:1187 PCCB:1370 SLC36A2:1506 SLC6A20:1870
Hyperchloriduria	0.42751711	5	9.306e−04	5.276e−01	CLCNKA:3 CLCNKB:21 KCNJ1:188 BSND:1259 SLC12A1:3855 NA
HYPERGLYCINURIA (disorder)	0.25402741	15	6.596e−04	5.276e−01	ACADM:590 SLC6A19:1179 GCSH:1187 PCCB:1370 SLC36A2:1506 SLC6A20:1870
Increased urinary potassium	0.42751711	5	9.306e−04	5.276e−01	CLCNKA:3 CLCNKB:21 KCNJ1:188 BSND:1259 SLC12A1:3855 NA
Kidney Calculi	0.13523541	53	6.670e−04	5.276e−01	CLDN16:1 AGXT:27 SLCO1B1:38 SLC4A1:625 AMBP:678 IL6:778
Macular pigmentary changes	0.20394742	21	1.218e−03	5.276e−01	PROM1:41 GPR179:281 TRPM1:612 PDE6B:973 ABCA4:997 GRM6:1229
Disorder of the genitourinary system	0.34870932	7	1.399e−03	5.715e−01	PLEC:88 ITGB4:587 SFRP1:1132 ITGA6:1286 FSHR:1338 MED12:4886
Turner Syndrome, Male	−0.14565442	38	1.901e−03	7.170e−01	ZHX2:195 MAPK3:205 GH1:464 GPNMB:471 ACP1:649 MPZL1:846
Cholestasis	0.05770888	248	1.838e−03	7.170e−01	CYP17A1:11 ABCB11:35 SLCO1B1:38 SLC23A1:39 AFP:104 GGTLC3:198
Skin Manifestations	0.26836862	11	2.058e−03	7.473e−01	CYP1A1:87 PLEC:88 ACAD8:314 NT5E:1686 GJB2:1716 IL33:1724
Metabolic Bone Disorder	0.16480128	29	2.137e−03	7.485e−01	KDM4B:43 CYP27A1:231 GLB1:305 IL6:778 PTH:840 PHEX:1181
Progressive intrahepatic cholestasis (di	0.18685729	22	2.420e−03	7.910e−01	ABCB11:35 GGTLC3:198 SLC10A2:329 NR1H4:776 HSD17B6:846 GGTLC1:851
Rare Diseases	0.23436139	14	2.399e−03	7.910e−01	AGXT:27 ARG1:361 HGD:566 DNMT3B:1006 ABCC6:1019 CPS1:2488
Acute myeloid leukemia, minimal differen	0.14924084	18	2.841e−02	8.056e−01	ELL:53 ZMYND11:1032 CD34:1406 ABCB1:1466 CD19:2599 NCR1:2619
Alport Syndrome, Autosomal Recessive	−0.25368175	5	4.947e−02	8.056e−01	COL4A4:2130 COL4A3:2389 MPV17:3406 COL4A5:4398 MYH9:6184 NA
Amyotrophy, monomelic	0.26657205	5	3.899e−02	8.056e−01	SMN1:1667 SOD1:2199 SMN2:2406 CEP126:5232 CPLANE1:5702 NA
Anemia, Diamond−Blackfan	0.09184676	45	3.315e−02	8.056e−01	THPO:834 LPIN2:941 CD38:1127 RPL31:1321 CD34:1406 COL3A1:1451
Anemia, hereditary spherocytic hemolytic	0.35518450	5	5.949e−03	8.056e−01	EPB42:141 ANK1:429 SPTA1:547 SLC4A1:625 SPTB:9048 NA
Anemia, Sickle Cell	0.04416719	170	4.764e−02	8.056e−01	ICAM4:37 COMT:122 RHD:147 CRYZ:154 SCD:189 MUC4:202
ARTHROGRYPOSIS MULTIPLEX CONGENITA, DIST	0.17988976	9	6.167e−02	8.056e−01	WNK4:831 FBN2:1163 SLC12A3:1485 WNK1:2390 PIEZO2:2451 REN:5806
Asthma, Occupational	0.14966424	17	3.268e−02	8.056e−01	TENM3:273 TACR2:666 PTGS2:1953 PRKCA:2988 TNF:3235 NAT2:3822
Ataxias, Hereditary	0.04865996	150	4.023e−02	8.056e−01	RNF216:69 SYNE1:100 DNMT1:105 HEXB:124 CYP27A1:231 SNAP25:391
Brain Injuries, Focal	0.08699499	56	2.446e−02	8.056e−01	ALB:299 IL6:778 IL13RA2:1082 IL1R1:1194 GABRB3:1256 IRF1:1311

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0005332 abnormal amino acid	0.08470551	112	2.140e−03	2.261e−01	SLC6A6:9 BCAT2:47 GNMT:225 ARG1:361 DAO:483 HGD:566
MP0001666 abnormal nutrient absorption	0.15136006	40	9.582e−04	2.261e−01	CLPS:15 SLC23A1:39 PNLIP:67 ABCG8:232 PNLIPRP2:242 ALB:299
MP0001186 pigmentation phenotype	−0.08533341	110	2.171e−03	2.261e−01	EDARADD:42 TYRP1:192 EDNRB:363 RB1:433 RLBP1:717 SUFU:804
MP0005395 other phenotype	0.09850132	83	2.053e−03	2.261e−01	AGXT:27 SLC23A1:39 NLRP5:106 DST:138 SCNN1G:168 PKHD1:311
MP0000372 irregular coat pigmentation	−0.23480228	14	2.375e−03	2.261e−01	NTRK1:102 SLC45A2:1004 HTR2C:1205 PMEL:1453 MCOLN3:1641 NSDHL:3596
MP0004043 abnormal pH regulation	0.22759365	13	4.530e−03	3.101e−01	RHCG:91 KCNJ1:188 SCNN1B:469 WNK4:831 HIF1AN:1452 SLC4A7:2238
MP0004811 abnormal neuron physiology	−0.05074751	256	6.153e−03	3.101e−01	PINK1:50 SCYL1:81 GNPAT:99 AMPH:161 DAG1:165 COL4A1:209
MP0002139 abnormal hepatobiliary system	0.27695195	8	6.706e−03	3.101e−01	ALDH1A1:127 ABCG8:232 FABP1:741 SLC22A1:1144 HNF1B:4424 ABCB4:5845
MP0002136 abnormal kidney physiology	0.04946518	265	6.642e−03	3.101e−01	CLDN16:1 CLCNKA:3 MAN2A1:19 AGXT:27 SLC23A1:39 BCAT2:47
MP0000001 mammalian phenotype	−0.02212318	2775	7.064e−03	3.101e−01	IL5RA:3 E2F8:6 AVPR2:7 PGGT1B:19 PSMB9:20 PPARGC1A:21
MP0005410 abnormal fertilization	0.10432730	56	7.166e−03	3.101e−01	ADAM2:20 KCNU1:81 HEXB:124 B4GALT1:129 PLCD4:236 ADAD1:276
MP0001765 abnormal ion homeostasis	0.07189967	99	1.415e−02	3.369e−01	CLDN16:1 CYP11A1:5 TRPM6:17 RHCG:91 SCNN1G:168 SLC26A3:172
MP0003195 calcinosis	0.25983483	8	1.097e−02	3.369e−01	AGXT:27 ABCC6:1019 FSHR:1338 CASR:3573 SLC12A1:3855 ENPP1:4291
MP0002168 other aberrant phenotype	0.10417261	51	1.036e−02	3.369e−01	AGXT:27 DST:138 SCNN1G:168 PKHD1:311 CPT1A:671 ST14:849
MP0002169 no abnormal phenotype	−0.02491522	1075	1.217e−02	3.369e−01	E2F8:6 GSDMD:24 OCRL:26 CITED2:68 ALDH3A1:86 HAS2:93
MP0002138 abnormal hepatobiliary system	0.25070070	8	1.412e−02	3.369e−01	PKHD1:311 ARG1:361 RELB:2581 HNF1B:4424 SOCS2:4512 MAP3K14:4980
MP0001756 abnormal urination	0.08345959	78	1.130e−02	3.369e−01	CLCNKA:3 BCAT2:47 DST:138 SCNN1G:168 KCNJ1:188 ALB:299
MP0002249 abnormal larynx morphology	−0.12866245	31	1.338e−02	3.369e−01	BMP7:528 ZIC3:1030 FBLN1:1328 HOXB3:1492 RXRA:1606 TBX1:2298
MP0000432 abnormal head morphology	−0.04524739	285	1.019e−02	3.369e−01	EDARADD:42 EP300:46 EVC:72 WNT9B:92 VAX1:164 DAG1:165
MP0002873 normal phenotype	−0.02469482	1080	1.280e−02	3.369e−01	E2F8:6 GSDMD:24 OCRL:26 CITED2:68 ALDH3A1:86 HAS2:93
MP0002184 abnormal innervation	−0.06236699	128	1.584e−02	3.591e−01	ERBB4:40 SCYL1:81 EPHA5:262 NGFR:397 FGFR3:426 ALCAM:669
MP0003755 abnormal palate morphology	−0.05603375	155	1.738e−02	3.760e−01	WNT9B:92 VAX1:164 PDS5B:285 HSPG2:313 EDNRB:363 FGFR3:426
MP0002080 prenatal lethality	−0.02249215	1198	1.869e−02	3.846e−01	SMARCC1:15 ERBB4:40 EP300:46 APOH:49 HMGN1:52 CITED2:68
MP0000371 diluted coat color	−0.09789354	48	1.939e−02	3.846e−01	TYRP1:192 MYO5A:843 SLC45A2:1004 FZD4:1308 SPAG9:1310 RXRA:1606
MP0009931 abnormal skin appearance	−0.04017721	290	2.144e−02	3.980e−01	AVPR2:7 DGAT1:33 EDARADD:42 NTRK1:102 PPP1R15B:162 SPINK5:219
MP0000653 abnormal sex gland	0.03267492	452	2.174e−02	3.980e−01	CYP11A1:5 PIWIL1:33 SULT1E1:44 AFP:104 HEXB:124 B4GALT1:129
MP0003011 delayed dark adaptation	−0.20440617	10	2.531e−02	4.433e−01	RLBP1:717 RDH11:1449 RXRA:1606 RPE65:2393 RDH5:2826 RDH8:4762
MP0000778 abnormal nervous system	−0.06521166	99	2.607e−02	4.433e−01	ARHGAP5:44 VAX1:164 NCK1:335 MAP3K12:685 MSX1:760 PDHA1:773
MP0005360 urolithiasis	0.22414915	8	2.822e−02	4.477e−01	AGXT:27 SLC7A9:774 SLC2A9:1340 SLC26A6:3639 SLC26A1:6026 OTC:6232
MP0002152 abnormal brain morphology	−0.02365554	868	2.819e−02	4.477e−01	SMARCC1:15 PPARGC1A:21 ERBB4:40 EDARADD:42 ARHGAP5:44 EP300:46
MP0001929 abnormal gametogenesis	0.03263516	404	2.938e−02	4.511e−01	CYP11A1:5 CYP17A1:11 PIWIL1:33 B4GALT1:129 FANCM:156 MOV10L1:170
MP0005085 abnormal gallbladder physiolo	0.12554395	24	3.357e−02	4.842e−01	ABCB11:35 CYP27A1:231 ABCG8:232 SLC27A5:257 SLC10A2:329 CYP7A1:690
MP0003718 maternal effect	0.10918326	32	3.297e−02	4.842e−01	SLC23A1:39 NLRP5:106 ZAR1:702 GPR3:818 TMPRSS6:1133 PHEX:1181
MP0005381 digestive/alimentary phenotyp	0.06006533	105	3.491e−02	4.887e−01	SLC26A3:172 SH3BP2:291 SCNN1B:469 GDF1:752 GLG1:795 SMARCAD1:807
MP0003698 abnormal male reproductive	0.02947723	453	3.823e−02	5.199e−01	CYP17A1:11 ADAM2:20 PIWIL1:33 SULT1E1:44 CHDH:72 KCNU1:81
MP0009643 abnormal urine homeostasis	0.04113668	216	4.052e−02	5.219e−01	CLDN16:1 CLCNKA:3 MAN2B1:4 MAN2A1:19 AGXT:27 GCDH:34
MP0001324 abnormal eye pigmentation	−0.06623324	81	4.056e−02	5.219e−01	GNPAT:99 UNC119:184 TYRP1:192 EDNRB:363 GPNMB:471 BMP7:528
MP0005551 abnormal eye electrophysiolog	0.05072604	134	4.480e−02	5.456e−01	CPLX3:26 PROM1:41 EGFLAM:112 MDM1:272 NPTX1:316 CACNB2:364
MP0002932 abnormal joint morphology	−0.05917119	98	4.454e−02	5.456e−01	HAS2:93 SHARPIN:222 HSPG2:313 ZAP70:323 PTCH1:339 BMP7:528
MP0006054 spinal hemorrhage	−0.20397898	8	4.585e−02	5.456e−01	C1GALT1C1:928 SPHK1:1190 FBLN1:1328 PSEN1:3158 FLI1:5462 C1GALT1:6444

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
testis	0.03168579	1329	3.332e−04	8.996e−03	BNC1:25 TIGD4:30 PIWIL1:33 FBXO24:42 CATSPERE:57 CFAP20DC:59
eye.development	−0.14573142	55	1.927e−04	8.996e−03	CITED2:68 ALDH3A1:86 VAX1:164 CYP1B1:404 BMP7:528 MSX1:760
liver	0.05738768	320	5.184e−04	9.331e−03	EHHADH:22 AGXT:27 ABCB11:35 SLCO1B1:38 AFP:104 IDO2:131
peyers.patch	0.06181887	163	6.917e−03	9.338e−02	SLC23A1:39 SI:83 SLC5A12:96 TRPM5:109 KRT20:115 B3GALT5:121
caudate.nucleus	−0.13623122	26	1.633e−02	1.009e−01	VAX1:164 FAM237A:249 SLC35D3:332 RLBP1:717 NR2E1:799 GABRG1:929
hippocampus.proper	−0.24710460	8	1.554e−02	1.009e−01	OLIG2:1157 NEUROG3:1474 FEZF2:1848 NTS:3608 GRP:4161 NEUROD2:4362
hypothalamus	−0.09366763	57	1.472e−02	1.009e−01	PCDH19:441 NGB:489 GSX1:718 PNMA8C:1296 ENKUR:1376 SNCG:1610
thyroid	0.05904137	144	1.518e−02	1.009e−01	CLCNKA:3 TLE6:61 IDO2:131 SORD:165 SLC26A4:292 TMC4:307
transformed.skin.fibroblast	−0.04526752	241	1.682e−02	1.009e−01	PLOD2:37 CITED2:68 EVC:72 HAS2:93 MYO1B:95 GPX8:231
cortex.kidney	0.06384008	108	2.262e−02	1.222e−01	CLDN16:1 CLCNKB:21 ENAM:82 SLC5A12:96 KCNJ1:188 NPHS1:191
pituitary.gland	−0.04437291	200	3.224e−02	1.583e−01	IL5RA:3 SLC38A8:163 PRLHR:182 TMIE:218 BAIAP3:254 RAB3B:304
cerebellum	−0.03292191	349	3.800e−02	1.710e−01	NRG2:45 TRIM3:101 CHRNB3:114 GPLD1:159 NXPH4:187 VWA5B2:193
transverse.colon	0.04592513	144	5.895e−02	2.449e−01	TRPM6:17 KRT20:115 SLC26A3:172 MUC4:202 SULT1B1:249 SLC17A4:405
Brodmann.area.9	−0.04236056	158	6.834e−02	2.460e−01	HCN1:176 EPHA5:262 SYT16:331 ERC2:371 CDH9:375 PHYHIPL:431
cerebellar.hemisphere	−0.02599353	453	6.413e−02	2.460e−01	NRG2:45 TRIM3:101 CHRNB3:114 SRGAP2B:139 AMPH:161 SSTR2:210
aorta	−0.03405767	150	1.531e−01	4.350e−01	GLIPR1:16 C1QTNF2:200 PPP1R14A:206 LTBP1:207 LMCD1:272 LTBP4:398
nucleus.accumbens	−0.04539307	91	1.363e−01	4.350e−01	VAX1:164 MINDY4B:197 FAM237A:249 GABRA5:751 RGS8:1014 ELMOD1:1085
substantia.nigra	−0.12711842	11	1.445e−01	4.350e−01	KLHL1:755 CHRM5:2042 RPE65:2393 SPX:2598 KCNE5:3887 TH:4616
tibial.nerve	−0.03149757	181	1.476e−01	4.350e−01	ACKR2:2 LGI2:199 BIRC7:311 NGFR:397 CYP1B1:404 ENTHD1:533
minor.salivary.gland	0.04839656	69	1.659e−01	4.480e−01	BPIFB1:2 PROM1:41 CRACR2B:250 ENPP3:644 PRSS8:742 DNASE2B:1178
EBV.lymphocyte	0.01598756	647	1.803e−01	4.637e−01	TMEM131L:24 SMC4:75 IQGAP3:86 ESCO2:113 MIS18BP1:119 KIF18B:135
coronary.artery	−0.10100413	14	1.910e−01	4.687e−01	COL4A1:209 ST6GALNAC5:342 CCDC190:702 CCDC3:882 PCOLCE2:2590 SOST:2682
spleen	−0.02219130	287	2.025e−01	4.755e−01	GSDMD:24 KAZALD1:147 IRAK3:175 ZAP70:323 CD2:373 TXK:515
C1.spinal.cord	−0.02572890	166	2.567e−01	5.332e−01	UGT8:140 SNORC:143 PKP4:328 PEX5L:638 HHIP:686 OR2K2:753
stomach	0.03754568	83	2.390e−01	5.332e−01	MYH13:10 TMED6:63 TMPRSS2:110 PGC:144 PDILT:641 EPN3:744
vagina	0.03504420	89	2.553e−01	5.332e−01	BNC1:25 SULT1E1:44 RHCG:91 SCNN1G:168 ADGRF4:376 EPHX3:920
leg.skin	0.02063259	208	3.102e−01	6.071e−01	BNC1:25 SULT1E1:44 CYP1A1:87 ANKRD35:102 ACER1:117 SPTLC3:140
suprapubic.skin	−0.02192199	180	3.148e−01	6.071e−01	EDARADD:42 ALOXE3:96 XCR1:157 LY6D:178 SPINK5:219 A2ML1:241
esophagus.mucosa	0.01953896	213	3.310e−01	6.163e−01	BNC1:25 CRB3:73 RHCG:91 ADGRF1:180 SLC44A5:182 ADGRF4:376
subcutaneous.adipose	−0.03254063	70	3.482e−01	6.267e−01	AVPR2:7 PNPLA2:467 ACVR1C:762 PKD1L2:839 GPAM:892 SLC19A3:983
blood	0.01601838	266	3.751e−01	6.533e−01	TREML2:6 SLC6A6:9 IL1R2:14 GP9:98 EPB42:141 MEFV:197
left.ventricle	0.02885002	67	4.157e−01	6.653e−01	TNNI3K:230 FSD2:426 MYOM2:684 ACTC1:689 CKMT2:966 MLIP:1040
pancreas	0.02544706	93	3.986e−01	6.653e−01	CPB1:7 CLPS:15 KCNK16:50 TMED6:63 PNLIP:67 CELA3B:118
uterus	−0.03218028	53	4.189e−01	6.653e−01	CAPN6:529 DCHS1:675 PTGER3:942 SULF2:1006 RSPO1:1236 TGFB1I1:1253
endocervix	0.02643412	69	4.492e−01	6.931e−01	ANGPT4:173 METTL27:224 ADGRA2:1087 SFRP1:1132 PWP2:1220 BDKRB1:1304
tibial.artery	−0.02017003	101	4.859e−01	7.288e−01	LDLRAD2:448 ACADL:883 MAP3K7CL:1041 NTRK3:1054 GPC6:1113 INHBA:1142
prostate	0.03089024	38	5.107e−01	7.454e−01	TTC6:349 SHROOM1:387 P2RX2:1004 CPAMD8:1237 HOXB13:1308 SCUBE2:1390
amygdala	−0.05924810	7	5.873e−01	7.854e−01	SLC38A8:163 BCAN:4191 NEUROD2:4362 SLC17A7:5348 TNR:7691 PCDH15:11267
Brodmann.area.24	−0.02488585	28	6.490e−01	7.854e−01	DOK6:1 TMEM271:1531 IDS:1817 TAC3:2708 NPTXR:3658 FAM189A1:3728
ectocervix	−0.03241237	25	5.753e−01	7.854e−01	TRHDE:605 COL5A1:1596 PRLR:2719 CPXM1:3674 FGF7:3928 HOXD13:4314